



TOUCH Switch

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Touch switches are used in many lamps and wall switches that have a metal exterior as well as on public computer terminals. A touchscreen includes an array of touch switches on a display. Basically, a touch switch is the simplest kind of tactile sensor. Described here is a touch switch that operates only when you touch two conductor terminals (say, A and B) simultaneously with your fingertip.

Circuit and working

Circuit diagram of a simple touch switch is shown in Fig. 1. It is built around four BC547 transistors (T1 through T4), a 5V single changeover relay (RL1), two 1N4007 diodes (D1 and D2) and two touch plates (TP1 and TP2). The circuit works off 5V DC.

Working of the circuit is straightforward. When TP1 is touched, the base voltages of transistors T1 and

T4 go high. As a result, relay RL1 energises to activate the load (light or fan) through relay contacts.

When TP2 is touched, the base voltage of transistor T2 goes high and transistor T3 conducts. The voltage developed across

resistor R2 de-energises relay RL1 to de-activate the load.

Touch plates TP1 and TP2 can be made from copper wires or copper plates.

Construction and testing

An actual-size PCB layout for the touch switch is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit, enclose it in a suitable box. Affix the touch plates on the front panel of the box such that you can touch them easily. **EFY**



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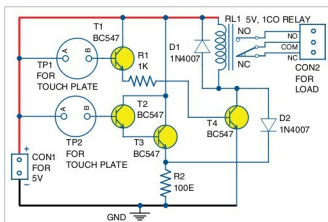


Fig. 1: Circuit diagram of the touch switch

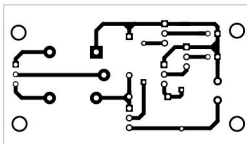


Fig. 2: Actual-size PCB layout of the touch switch

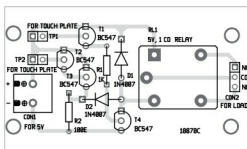


Fig. 3: Components side of the PCB

PARTS LIST

Semiconductors:

- T1-T4 - BC547 npn transistor
- D1, D2 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-kilo-ohm
- R2 - 100-ohm

Miscellaneous:

- CON1 - 2-pin terminal connector
- TP1, TP2 - Touch plate
- RL1 - 5V, 1CO PCB-mount relay
- 5V power supply

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